

AQA (Trilogy) Combined Science GCSE

Unit 6.3 Particle Model of Matter

Test (Levels 4 – 9)

Time allowed: **50 minutes**

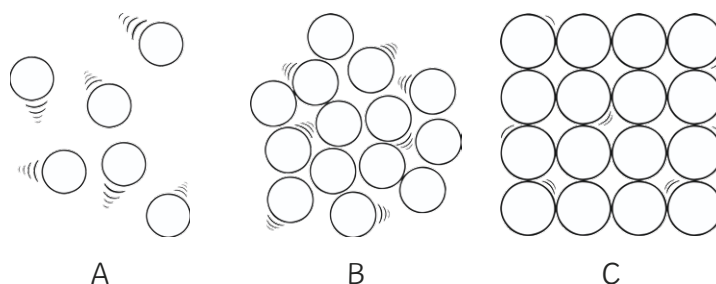
Question	Links to Student Progress Sheet	Score	Total Marks Available
1	6.3.1.1.a b 6.3.3.1.c		7
2	6.3.1.1.d 6.3.3.1.a, b d		4
3	6.3.1.2 6.3.2.3.d f		14
4	6.3.1.2.a 6.3.2.1.a 6.3.2.3.e		8
5	6.3.2.3 d 6.3.3.1.a		11
6	6.3.1.1		6
Total	6.3		50

Score	Estimated Grade
47 – 50	9
40 – 46	8
33 – 39	7
28 – 32	6
24 – 27	5
20 – 23	4
0 – 19	U

Please be aware that these grade boundaries are Twinkl's own, and may not be in-line with AQA's own.

Area for Improvement:

1. The boxes show three states of matter as simple models.



a. What equation is used to correctly calculate the density of a material?

[1 marks]

b. There are limitations to the particle model shown above. State two such limitations.

[2 marks]

c. Explain why the pressure inside a container increases as the gas inside has an increase in temperature. The container has a fixed volume.

[4 marks]

7

2. The particle model is one way of explaining density.

a. What unit is density (ρ) measured in?

kilograms, kg

☐

kilograms per metre cubed, kg/m^3

☐

joules, J

☐

joules per kilogram, J/kg

☐

[1 mark]

b. Use the answers in the box to complete the following sentences:

The particles in a gas move in a random way; they are _____ moving. The particles in a gas have more _____ energy if temperature is increased.

kinetic

thermal

nuclear

constantly

rapidly

[2 mark]

c. If a gas in a container is heated, what happens to the pressure inside?

[1 mark]

4

- 3.
- a. Describe what the phrase 'specific latent heat of vaporisation' means.

[2 mark]

- b. Three different states of matter are gas, liquid and solid. Describe how gas and solid states are different in terms of particle arrangement and movement.

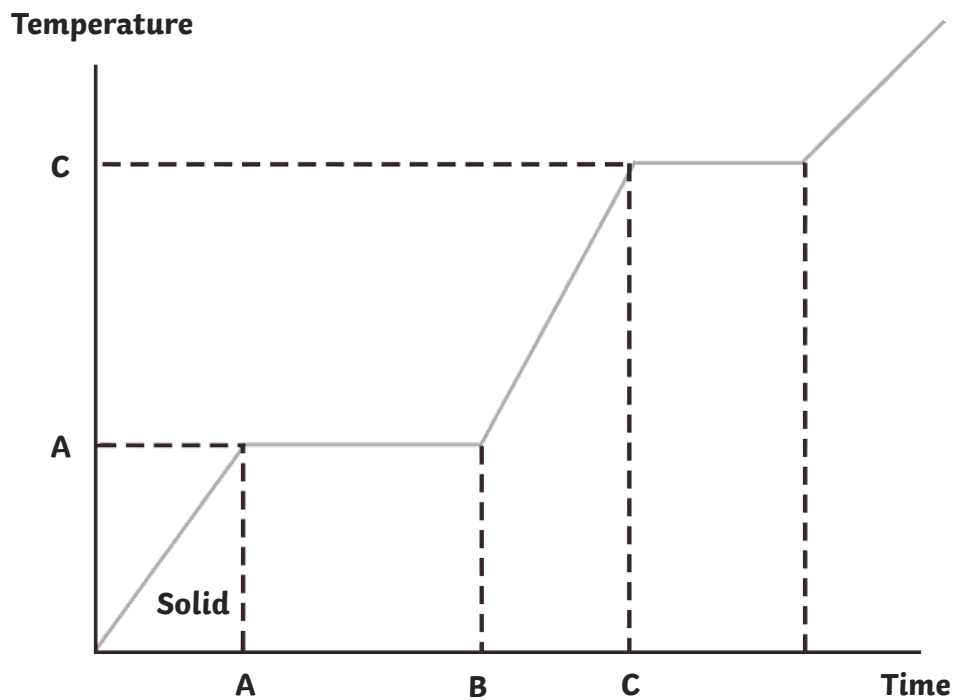
[4 marks]

- c. Calculate the amount of energy that is needed for a kettle to boil when 0.021kg of water changes to steam. The specific latent heat of vapourisation of water is 2.26×10^6 J/kg.

Answer: Energy required = _____ J

[2 marks]

- d. As a substance is heated, the temperature will change. The graph below shows different stages of the heating process. (not to scale)



Identify: A _____

[1 mark]

B _____

[1 mark]

Describe what is happening in section:

AB _____

BC _____

[4 marks]

14

4. Car windscreens freeze over in low temperatures. The windscreen has a heating element in the glass to defrost the ice which has settled on it.
- a. The element can supply energy to the ice without the temperature of the ice increasing. What is the name of this?

[1 mark]

- b. When investigating different heating elements, what control variable might a car manufacturer use?

[1 mark]

- c. Explain how the element changes the particle arrangement as the ice is melting and how this is different to the particle arrangements in other states of matter. You may also draw a diagram below to help your explanation.

[6 mark]

8

5. Properties of gases can be explained using the particle model.

a. How do particles move in a gas?

[1 mark]

b. At 100 °C, water can exist as a gas or a liquid. At equal masses, gaseous water at 100 °C has more energy than liquid water at the same temperature. Explain why.

[2 marks]

c. The particles in a gas will change speed when heated. Explain why.

[3 marks]

d. When a gas cools, it changes state and condensation can form.

The density of water is 1000 kg/m³

Specific latent heat of vapourisation of water = 2.26×10^6 J/kg.

If a volume of 3.1×10^{-5} m³ condensation forms on a mirror, calculate the energy released when condensation is formed.

Answer: Energy required = _____ J

[5 marks]

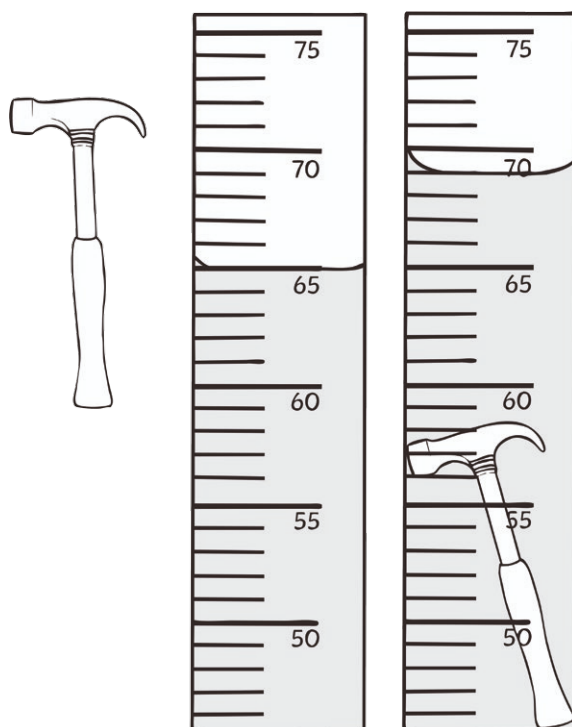
11

6.

- a. How do you find the density of an irregularly shaped object? Write a method and include the apparatus you would use.

[5 marks]

- b. Look at the diagram below. By how much has the water been displaced?

[1 mark]**5**

End of Test